

Ning-Hsu (Albert) Wang

☎ +886-922-105-719 ✉ albert.nhwang@gmail.com
🌐 [linkedin.com/in/ning-hsu-albert-wang](https://www.linkedin.com/in/ning-hsu-albert-wang) 🌐 [albert100121.github.io](https://github.com/albert100121)

Research Interest

Computer Vision, Machine Learning, 3D Geometry & Reconstruction, VR/AR, 360° Vision, Computational Photography, Robotic Perception, Generative AI

EDUCATION BACKGROUND

National Tsing Hua University

Master of Science in Electrical Engineering, advised by Prof. Min Sun

GPA: 4.3/4.3

Hsinchu, Taiwan

Feb. 2018 – Aug. 2020

National Chiao Tung University

Bachelor of Science in Mechanical Engineering

Last 60 credits GPA: 3.74/4.3

Hsinchu, Taiwan

Sep. 2013 – Jun. 2017

PUBLICATIONS

Ning-Hsu Wang, Yu-Lun Liu, “Depth Anywhere: Enhancing 360 Monocular Depth Estimation via Perspective Distillation and Unlabeled Data Augmentation”, Neurips 2024. [\[link\]](#)

Ning-Hsu Wang, Ren Wang, Yu-Lun Liu, Yu-Hao Huang, Yu-Lin Chang, Chia-Ping Chen, Kevin Jou, “Bridging Unsupervised and Supervised Depth from Focus via All-in-Focus Supervision”, ICCV 2021. [\[link\]](#)

Cheng Sun, Chi-Wei Hsiao, **Ning-Hsu Wang**, Min Sun, Hwann-Tzong Chen, “Indoor Panorama Planar 3D Reconstruction via Divide and Conquer”, CVPR 2021 Oral. [\[link\]](#)

Ning-Hsu Wang, Bolivar Solarte, Yi-Hsuan Tsai, Wei-Chen Chiu, Min Sun, “360SD-Net: 360° Stereo Depth Estimation with Learnable Cost Volume”, ICCV-W 2019 Spotlight, ICRA 2020. [\[link\]](#)

PATENT

Methods and Apparatuses of Depth Estimation from Focus Information

Ren Wang, Yu-Lun Liu, Yu-Hao Huang, Ning-Hsu Wang

U.S. Patent no. 11,967,096, filed Feb. 2022, published Sep. 2022, issued Apr. 2024. [\[link\]](#)

MediaTek

Feb. 2022

EXPERIENCE

Google

Image Processing and Machine Learning Engineer

Taipei, Taiwan

Jun. 2024–Present

- Designed and integrated next-generation ML features for edge devices, with emphasis on computational photography and image signal processing (ISP) algorithms.
- Proposed ML solutions based on product requirements and optimized image/video models for real-time performance on portable hardware.
- Developed end-to-end pipelines integrating ML algorithms with ISP modules, in close collaboration with cross-functional teams.
- Oversaw performance evaluation and integration of ML features across the product portfolio.

Taiwan AILabs, Metaverse Team

Machine Learning Research Engineer

Taipei, Taiwan

Aug. 2021–Apr. 2024

- **Metaverse Project:**
 - Optimized **360 Video Face Detection and De-identification**, improving inference speed by **400%**.
 - Developed a novel **light-source representation** and built a neural network model and dataset for realistic **object insertion, lighting, and shadow casting**, delivered within six months.

- Integrated **EquiConv with RAFT optical flow** to enhance virtual scene walk-through stability and contributed to four additional features.
- Devised strategy for **multi-task Generative AI (stable-diffusion)** algorithm development focusing on style transfer, super resolution, and panorama image generation.
- **Video Enhancement Project:** Built a high-performance video enhancement pipeline combining **SOTA video decompression and super-resolution** methods, matching competitor performance within a month.
- **Avatar and Virtual Studio:** Designed algorithm pipelines for avatar movement and **human motion generation**, contributed to development efforts, and worked on background removal and replacement using **text-to-image Generative AI** models.

MediaTek, MM, MTD, IVP

Hsinchu, Taiwan

Research intern in Computer Vision

Feb. 2020-Feb. 2021

- Published an **IEEE conference paper** in Computer Vision (**ICCV 2021**) and obtained a **US Patent** in the field of depth estimation for images with bokeh effects (shallow depth of field).
- Introduced a novel dataset for depth estimation on blurred images and developed an innovative unsupervised training technique.
- Demonstrated exceptional performance by surpassing SOTA methods with significant margins on multiple datasets: DDFF-12-Scenes (5.5%), HCI-4D-Light-Field (20%), and Defocus-Net (27%).

VSLab, National Tsing Hua University

Hsinchu, Taiwan

Graduate Research Assistant

Feb. 2018-Aug. 2020

- Published **two IEEE conference papers** in the fields of Computer Vision and Robotics (**ICRA 2020, CVPR 2021 Oral**).
- Led the 360° Stereo Project, under the guidance of [Prof. Min Sun](#), [Prof. Wei-Chen Chiu](#), and [Dr. Yi-Hsuan Tsai](#); focused on planar reconstruction, co-advised by [Prof. Hwann-Tzong Chen](#).
- Presented two 360° stereo datasets and developed a novel deep neural network for depth estimation on 360° stereo images (ICRA 2020).
- Introduced a new benchmark and method utilizing indoor human structures and 360° images for indoor panorama planar reconstruction (CVPR 2021 Oral).

PROFESSIONAL ACTIVITY

Served as a reviewer for:

- Journals: RA-L, TPAMI, IJCV
- Conferences: AAAI 2023, CVPR 2023, ECCV 2024, ICCV 2025, Neurips 2025

SKILLS

Programming: Python, C/C++, HTML/CSS

Tools: PyTorch, TensorFlow, OpenCV, Scikit-Learn, Vim, Linux, Git, L^AT_EX

Software/Hardware: LabVIEW, Matlab, LTSpice, ANSYS-Fluent, AutoCAD, Solidworks, Arduino, 8051

Language: Mandarin (native), English (fluent, TOEIC: 900)

HONORS & AWARDS

Honorary Member of The Phi Tau Phi Scholastic Honor Society of the Republic of China

2020

Appier Conference Scholarship for Top Researches on Artificial Intelligence

2020

Arctic Code Vault Contributor (GitHub)

2020